# Armandt Erasmus

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## Education

**University of Pretoria** Graduation: December 2024
*Bachelor of Science in Physics - Second Major: Mathematics* *Pretoria, South Africa*
*Bachelor of Science Honours in Physics specialising in Financial Engineering* *Graduation: December 2026*

**Institute and Faculty of Actuaries**
*Actuarial Exams: CB1, CB2, CB3, CS1, CM1, CP3 - (Expected 2027: CS2, CM2, CP2)*

## Experience

**Actuarial Product Engineer** Dec 2026 – Present
*InsureAI* *Cape Town, South Africa*

- Engineer production-ready actuarial software by translating modelling research and methodologies into scalable Python packages, backend services, APIs and SDKs, enabling robust and practical insurance modelling workflows.

**Actuarial Analyst/Developer** Feb 2026 – Dec 2026
*Miway Insurance Limited – Operational Efficiencies, Research & Development* *Midrand, South Africa*

- Using Next.JS and Django, led development of a proprietary claims estimation platform incorporating GLMs, Normalised Cost of Repair (NCR), salvage and claims development methodologies to support pricing, cost estimation and reserving, deployed on AWS and delivering R24M+ in annual cost savings.
- Led the enterprise deployment of a document-forensics platform for insurance fraud detection, driving adoption across 3 business units and 75+ users, coordinating 14+ Manco stakeholders and replacing manual verification processes to deliver an estimated R7M annual benefit.
- Architected and engineered a 24-module forensic fraud engine, developing the application layer in Django, data infrastructure in PostgreSQL, and integrating probabilistic evidence scoring and automated tamper detection to quantify document-level fraud risk and generate investigator-ready assessments within seconds.

**Actuarial Analyst** Jul 2025 – Feb 2026
*SolveCo Actuarial Services (Pty) Ltd* *Pretoria, South Africa*

- Built a centralised multi-asset financial reporting engine (VBA) covering 50+ portfolios and 10+ asset classes, standardising valuations, fair-value movements, cash-flow timing, and FX attribution to deliver IFRS, US GAAP, and GIPS-aligned TWR/MWR performance reporting.
- Developed a Python-based model validation framework that automated data preparation and testing for long-term insurance pricing models, improving governance and auditability while reducing testing timelines by 50%.

**Machine Learning Engineer** Jul 2023 – Jul 2024
*Google Developer Student Clubs* *Pretoria, South Africa*

- Led applied machine learning initiatives at the University of Pretoria, developing and delivering end-to-end ML solutions in Python, from data preprocessing and feature engineering through model training, evaluation and deployment using Scikit-learn and Streamlit for real-world business use cases.

## Projects

**Claimly - Computer Vision Vehicle Damage Assessment Platform** | *Python, SQL, PyTorch, Django*

- Built an end-to-end ML-driven claims estimation platform that converts vehicle images into priced repair estimates, integrating vehicle-part segmentation, damage attribution and pricing in a Django and SQL backend.
- Deployed two YOLO11 instance-segmentation models covering 23 vehicle-part and 6 damage classes; developed mask-overlap attribution to map detected damage to specific panels and generate reviewable estimates in 5 seconds per image on CPU.

## Skills & Certifications

**Technical**: Python, Prophet, SQL (PostgreSQL, MySQL, PL/SQL), C++, R (Shiny), Excel, VBA, MATLAB, SAS

**AI/ML & Data**: Databricks, Scikit-Learn, TensorFlow, PyTorch, NumPy, Pandas, PySpark, Power BI, Scala

**Project Management**: Agile, Scrum, Stakeholder Management, Risk Management, Scheduling, Project Planning

**Software Engineering**: Django, React, Streamlit, Next.JS, REST APIs, Docker, Kubernetes, Git, Azure/AWS

**Certifications**: CFI Financial Modeling & Valuation Analyst (FMVA®), CFI Capital Markets & Securities Analyst (CMSA®), IFRS 17: Insurance Contracts - CIMA, IBM Data Science Professional, Google Project Management